# Application Layer

Week 10

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ACU
AUSTRALIAN CATHOLIC UNIVERSITY

# Chapter 10: Application Layer



## Network Basics

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# Chapter 10 Objectives

- Explain how the functions of the application layer, session layer, and presentation layer work together to provide network services to end user applications.
- Describe how common application layer protocols interact with end user applications.
- Describe, at a high level, common application layer protocols that provide Internet services to end-users, including WWW services and email.
- Describe application layer protocols that provide IP addressing services, including DNS and DHCP.
- Describe the features and operation of well-known application layer protocols that allow for file sharing services, including: FTP, File Sharing Services, SMB protocol.
- Explain how data is moved across the network, from opening an application to receiving data.


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# What are we learning today?

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What will we learn this week?

- What the Application Layer does in computer networks.
- How the Application, Presentation and Session layers work.
- Common protocols such as DNS, HTTP, HTTPS, FTP and SMTP.
- How client-server communication works.
- How peer-to-peer (P2P) networks work.


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# What are we learning today?

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Australian example: When a student opens the ACU website, sends an email or downloads learning materials, application-layer protocols help make these services possible.

Discussion question: Which network services do you use every day?

Simple answer: Websites, email, online learning platforms, video calls and file sharing.


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# What Is the Application Layer?

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The Application Layer is the top layer of the OSI model (Layer 7).

It provides network services to software applications.

Examples include:

- Opening websites.
- Sending and receiving emails.
- Transferring files.
- Looking up website addresses.


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# What Is the Application Layer?

Australian example: When you access www.acu.edu.au, your browser uses application-layer protocols to communicate with web services.

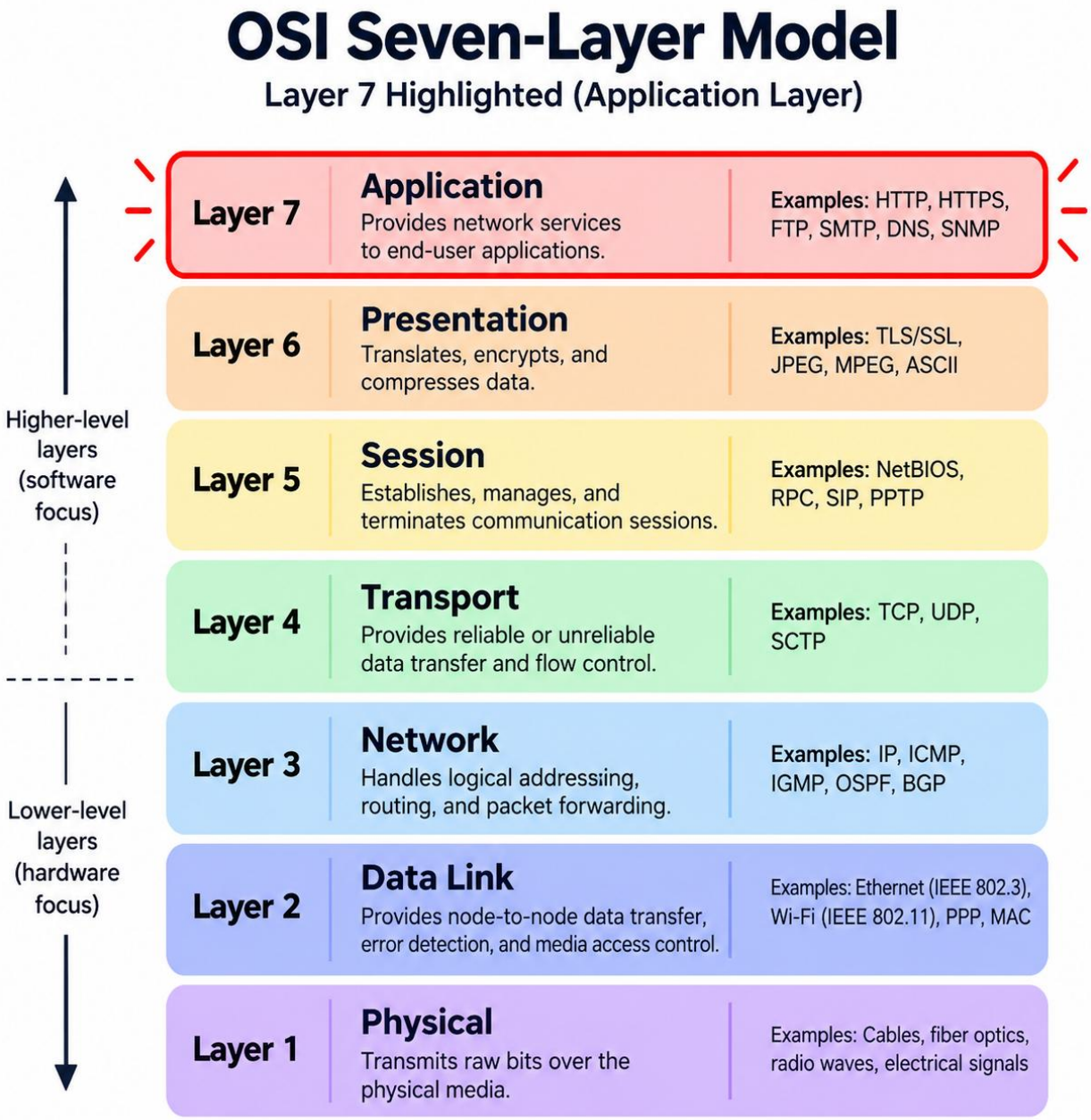
Discussion question: Does the Application Layer physically send signals through network cables?

Simple answer: No. It provides services to applications. Lower layers handle the movement of data across the network.


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# What Is the Application Layer?

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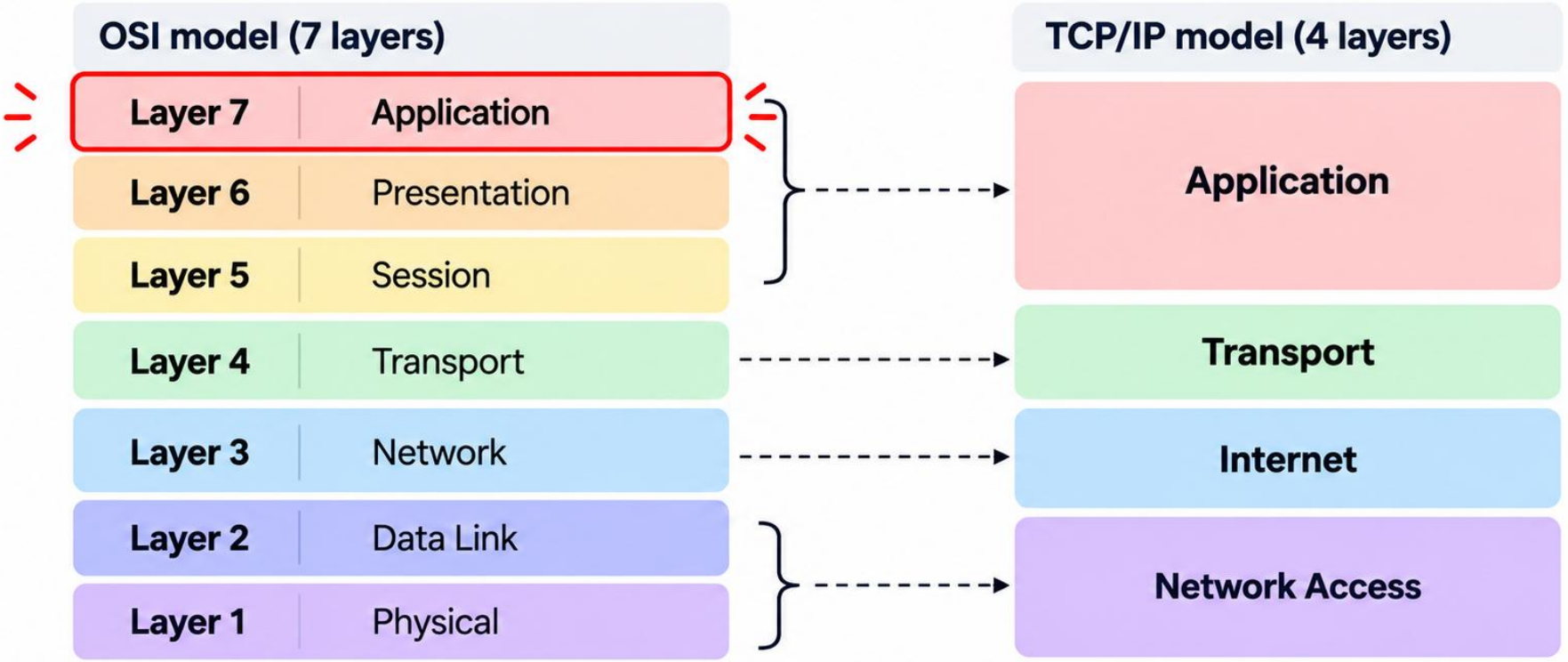




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# What Is the Application Layer?

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The TCP/IP Application Layer covers functions associated with the top three OSI layers.



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# What Is the Application Layer?

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Australian example: When a student joins an online class, application software uses network services while the lower layers carry data between devices.

Discussion question: How many OSI layers are represented within the TCP/IP Application Layer?

Simple answer: Three: Application, Presentation and Session.


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# What Does the Presentation Layer Do?

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The Presentation Layer is Layer 6 of the OSI model.

Its main functions include:

- Data conversion: Making data understandable to different systems.
- Compression: Reducing the size of data.
- Encryption and decryption: Protecting information during communication.


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# What Does the Presentation Layer Do?

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These are conceptual OSI functions; real TCP/IP applications may implement them through application protocols and supporting software.

Australian example: When students download compressed learning resources, compression helps reduce file sizes and download time.

**Discussion question:** Why is data compression useful?

Simple answer: It reduces the amount of data that needs to be stored or transmitted.


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# What Does the Session Layer Do?

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The Session Layer is Layer 5 of the OSI model.

It helps manage communication sessions between applications.

Its functions include:

- Starting a communication session.
- Maintaining communication.
- Coordinating the exchange of information.
- Ending or recovering a session when supported.


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# What Does the Session Layer Do?

Australian example: During an online tutorial between students in Sydney and Melbourne, the application must manage the ongoing communication session.

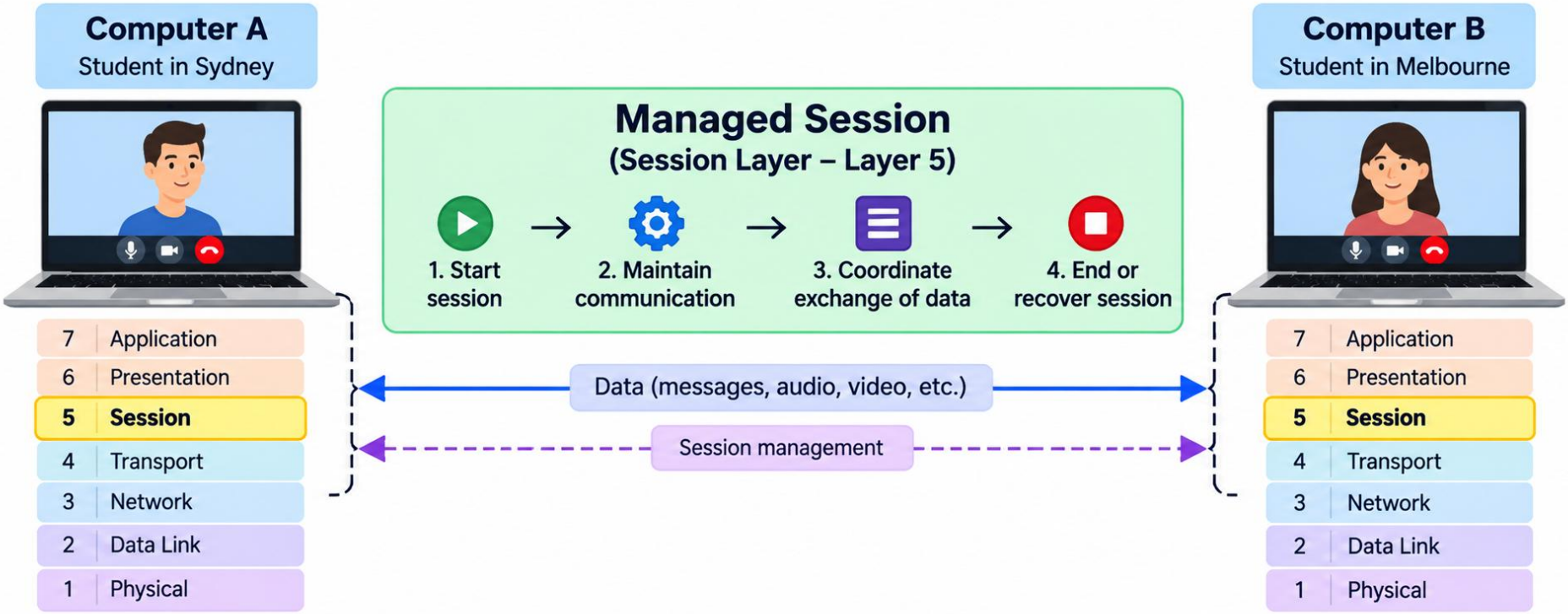
**Discussion question:** Why do applications need to manage sessions?

Simple answer: To keep communication organised and maintain the interaction between users or systems.


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# What Does the Session Layer Do?

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# Common Application Layer Protocols

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| Protocol | Main purpose |
|---|---|
| DNS | Finds IP addresses associated with domain names |
| HTTP | Transfers web resources |
| HTTPS | Secure web communication |
| FTP | Transfers files |
| TFTP | Simple file transfers |
| SMTP | Sends email |
| POP3 | Retrieves email |
| IMAP | Accesses and synchronises email |
| SSH | Secure remote access |
| Telnet | Unencrypted remote terminal access |


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# Common Application Layer Protocols

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Australian example: Accessing a university website involves DNS for name resolution and typically HTTPS for secure web communication.

Discussion question: Which protocol is commonly used to access a secure website?

Simple answer: HTTPS.


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# DNS: How Do We Find a Website?

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DNS stands for Domain Name System.

Computers communicate using IP addresses, but people usually remember website names.

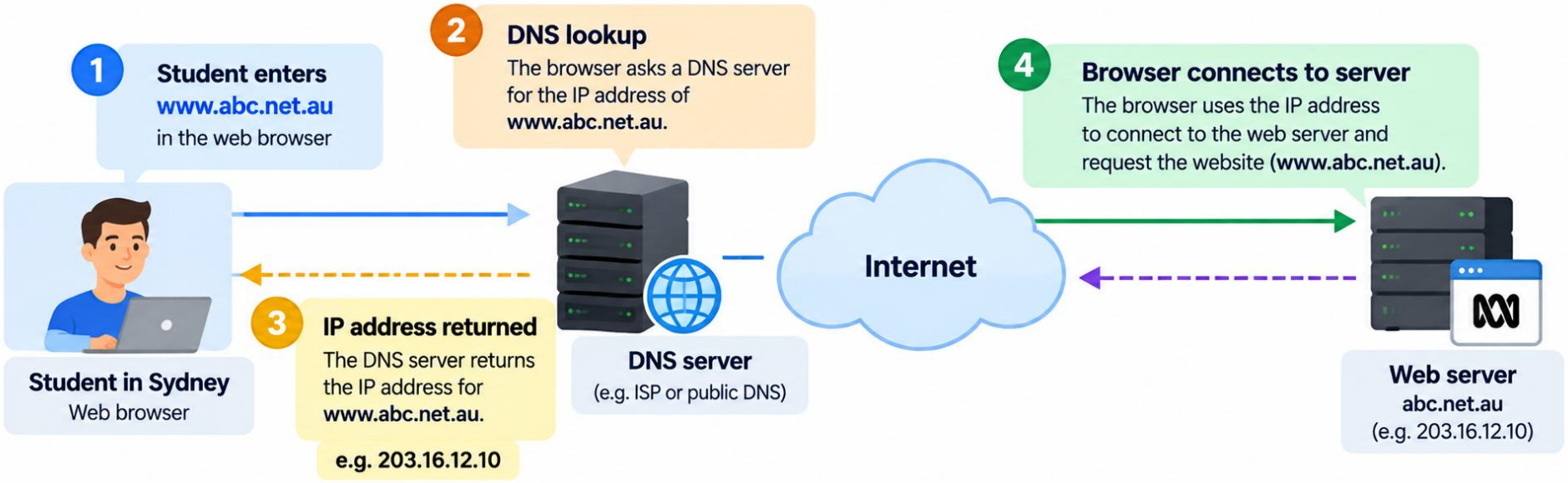
DNS helps translate domain names into IP addresses.

Example:


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# DNS: How Do We Find a Website?

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**Australian example:** A student in Sydney enters **www.abc.net.au** and DNS helps the browser find the appropriate server address.



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# DNS: How Do We Find a Website?

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**Discussion question: Why do we need DNS?**

Simple answer: Because website names are easier for people to remember than IP addresses.

Wireshark demonstration recommended: Capture a DNS query and response using the dns display filter.


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# HTTP and HTTPS: Accessing Websites

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HTTP stands for Hypertext Transfer Protocol.

It allows browsers and web servers to exchange web resources.

HTTPS is HTTP communication protected using TLS encryption.

Common ports:

- HTTP: TCP 80
- HTTPS: TCP 443 or HTTP/3 over QUIC using UDP 443


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# HTTP and HTTPS: Accessing Websites

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Australian example: When students sign in to an Australian university learning platform, HTTPS helps protect their login information during transmission.

Discussion question: Why should we use HTTPS rather than HTTP for login pages?

Simple answer: HTTPS encrypts the communication and helps protect sensitive information.

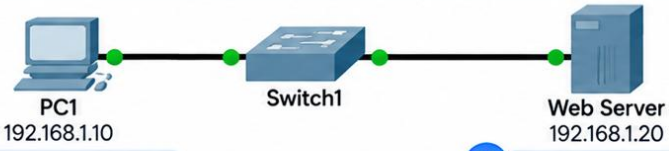

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# HTTP and HTTPS: Accessing Websites

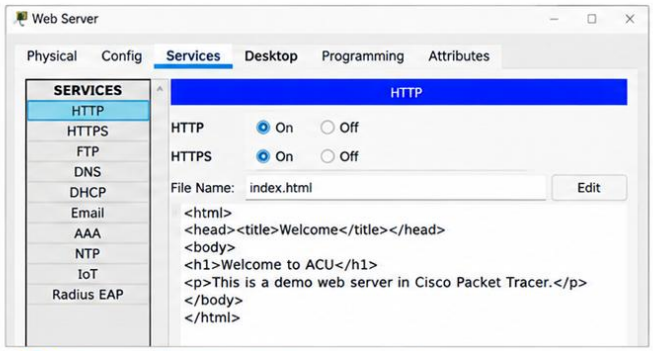
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## Cisco Packet Tracer Demonstration

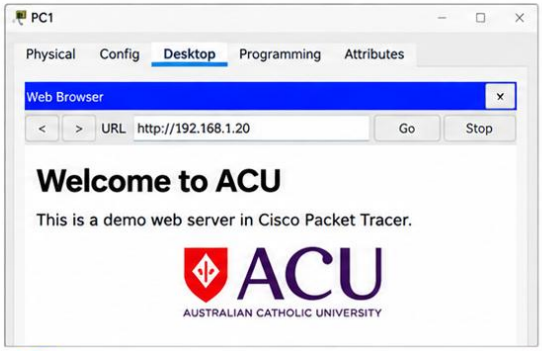
Configure a web server and access it from a PC using the browser




1. PC configured with an IP address
2. Web server configured with HTTP/HTTPS




3. Enable HTTP/HTTPS on the server and add a simple web page




4. On the PC, open the browser and enter the server IP address to access the web page




### Australian example

When students sign in to an Australian university learning platform, **HTTPS** helps protect their login information during transmission.

### Discussion question

Why should we use HTTPS rather than HTTP for login pages?

### Simple answer

HTTPS encrypts the communication and helps protect sensitive information.



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# Email Protocols: SMTP, POP3 and IMAP

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Email systems use different protocols for sending and receiving messages.

| Protocol | Function |
| --- | --- |
| SMTP | Sends and relays emails |
| POP3 | Downloads emails from a server |
| IMAP | Accesses and synchronises emails across devices |


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# Email Protocols: SMTP, POP3 and IMAP

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Australian example: A university staff member sends an email from a laptop and later reads messages on a mobile phone. IMAP can help keep the mailbox synchronised.

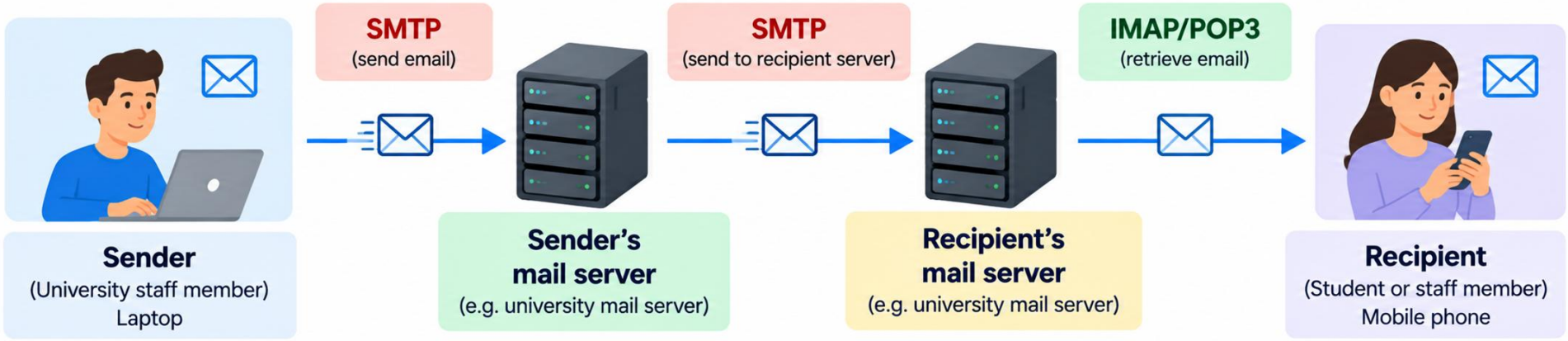
**Discussion question:** Which protocol is normally used to send an email?

Simple answer: SMTP.


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# Email Protocols: SMTP, POP3 and IMAP

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# What Is the Client-Server Model?

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In a client-server network:

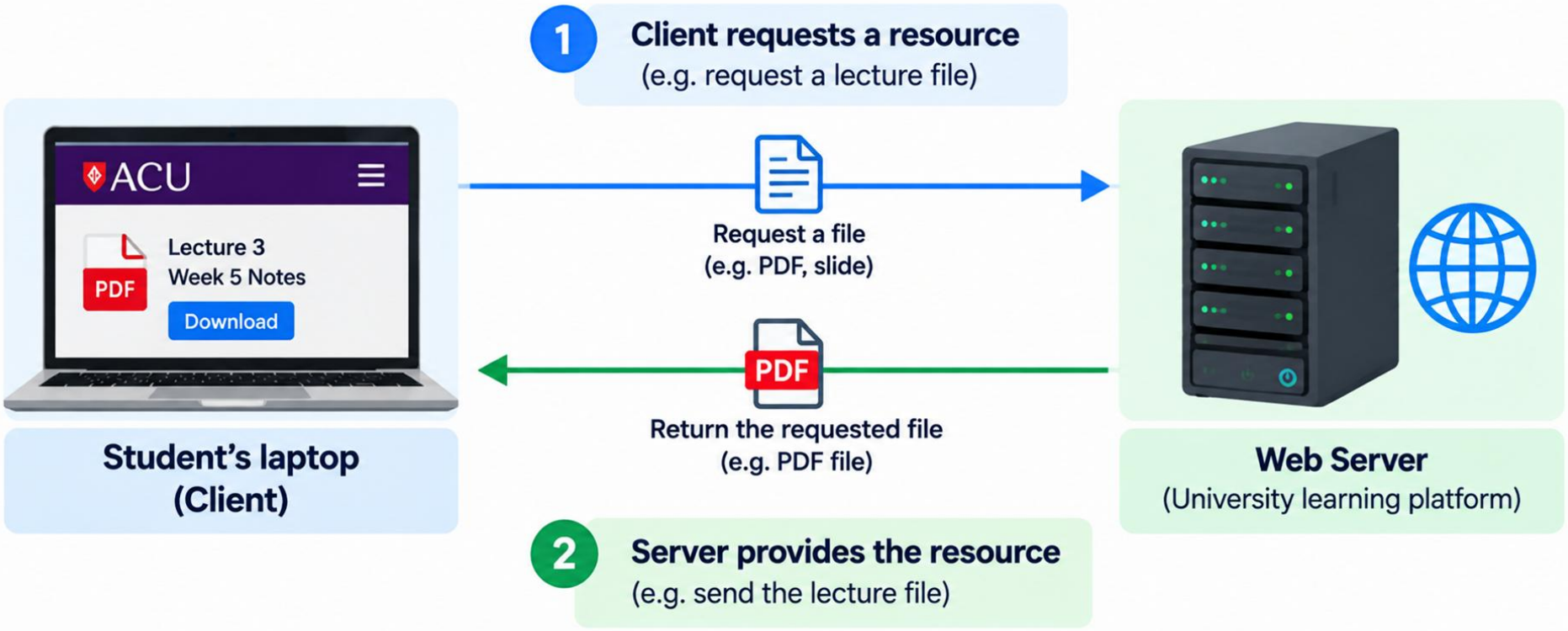
- The client requests a service or resource.
- The server provides the requested service or resource.
- Communication usually follows a request-response pattern.

Example:


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# What Is the Client-Server Model?

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# What Is the Client-Server Model?

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Australian example: When a student downloads a lecture file from an online learning platform, the student's device acts as a client and the platform's server provides the file.

**Discussion question:** When you open a university website, is your laptop usually the client or server?

Simple answer: The client.

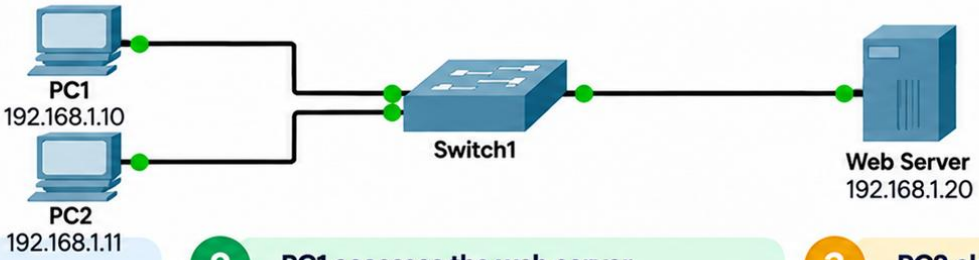

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# What Is the Client-Server Model?

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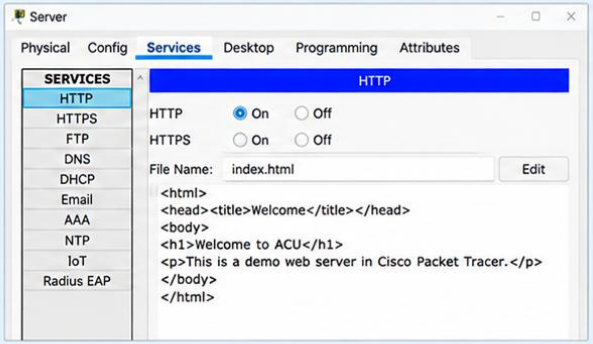
## Cisco Packet Tracer Demonstration

One server, one switch and two client PCs



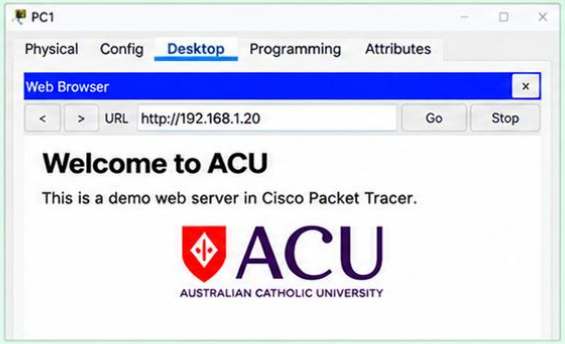

### 1 Web server configured

Enable HTTP and add a simple web page.



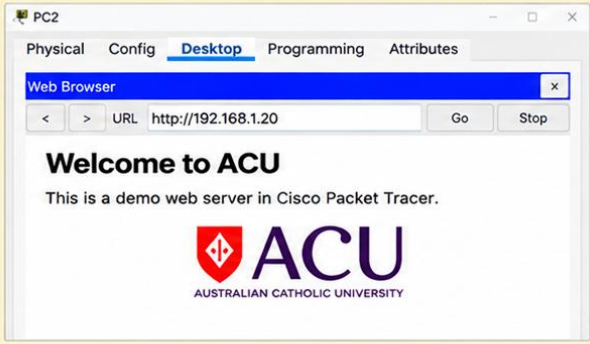

### 2 PC1 accesses the web server

Open the browser and enter the server IP address.




### 3 PC2 also accesses the web server

Open the browser and enter the same server IP address.






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# What Is Peer-to-Peer (P2P) Networking?

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In a peer-to-peer network, devices can share resources directly with other devices.

A peer may act as both a client and a server, depending on the request.

Examples include:

- Sharing files between computers.
- BitTorrent file distribution.
- Some decentralised applications.


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# What Is Peer-to-Peer (P2P) Networking?

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Australian example: Computers in a small Sydney office could share selected files directly over a local network rather than relying on a dedicated file server.

**Discussion question:** What is the main difference between client-server and P2P networking?

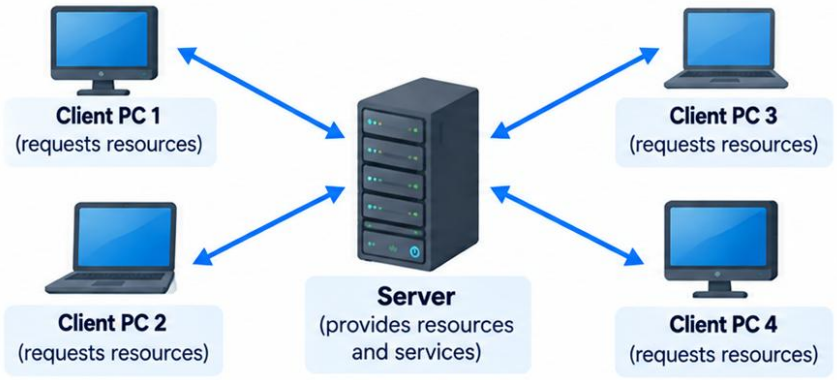
Simple answer: Client-server systems use designated servers to provide services. In P2P networks, devices can provide services directly to one another.


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# What Is Peer-to-Peer (P2P) Networking?

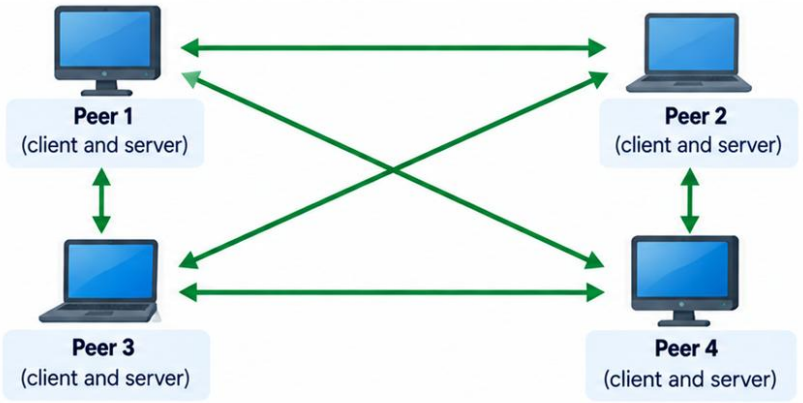
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## Client–Server Topology




- Clients request a service or resource from a central server.
- The server provides the requested service or resource.
- Communication follows a request–response pattern.
- Common in organisations, schools and universities.

## Peer-to-Peer (P2P) Topology




- Devices (peers) can share resources directly with each other.
- A peer may act as both a client and a server.
- No dedicated central server is required.
- Common in small offices, home networks and file-sharing systems (e.g. BitTorrent).


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# Client-Server vs Peer-to-Peer: Class Discussion

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| Feature | Client-server | Peer-to-peer |
|---|---|---|
| Main structure | Clients connect to servers | Peers communicate directly |
| Resource management | Often centralised | Often distributed |
| Administration | Easier to manage centrally | Can be harder to control |
| Example | University learning platform | BitTorrent file sharing |
| Scalability | Depends on server infrastructure | Depends on P2P design |


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# Client-Server vs Peer-to-Peer: Class Discussion

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Class activity) A small Australian business has five computers and wants to share files. A university has thousands of students who need secure access to learning resources.

**Discussion question:** Which networking model would you recommend for each organisation?

Simple answer: A small business may use P2P for simple file sharing. A university would normally use a centrally managed client-server system for security, access control and administration.


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# How Does HTTP Work?

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HTTP (Hypertext Transfer Protocol) allows a browser to request information from a web server.

- HTTP uses a request–response process.
- The client (browser) sends a request.
- The server processes the request.
- The server returns a response, such as a webpage, image or document.


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# How Does HTTP Work?

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Example in Australia:

A student opens the ACU website.

1. The browser requests a webpage.

2. The web server receives the request.

3. The server sends back the webpage.

4. The browser displays the content.

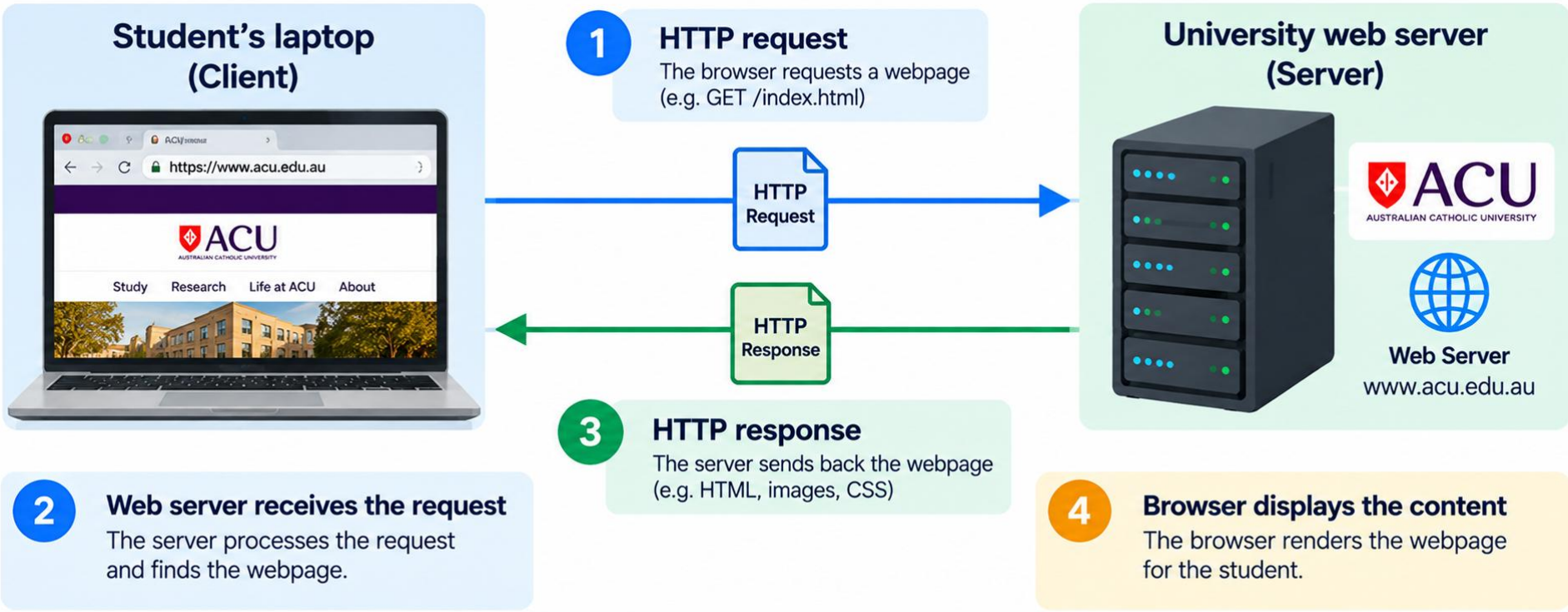
**Discussion question:** When you open a university website, who sends the first HTTP request?

Simple answer: Your browser (the client) sends the request to the web server.


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# How Does HTTP Work?

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# HTTP Methods: GET, POST and PUT

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HTTP uses different methods to tell the server what action is requested.

| Method | What it does | Example |
|---|---|---|
| GET | Requests information | Opening a webpage |
| POST | Submits information for processing | Submitting an online form |
| PUT | Creates or replaces a resource at a specified address | Updating a stored document through an API |


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# HTTP Methods: GET, POST and PUT

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Example in Australia:

- GET: A student opens an ACU course page.
- POST: A student submits a university registration form.
- PUT: A university application updates a stored record through an API.

**Discussion question:** A student clicks Submit on an online registration form. Which HTTP method is commonly used? Simple answer: POST, because information is being sent to the server for processing.


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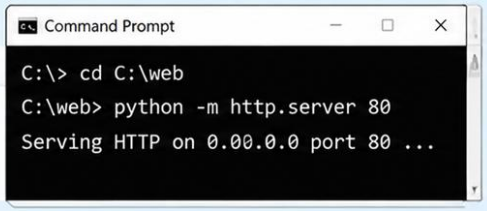
# HTTP Methods: GET, POST and PUT

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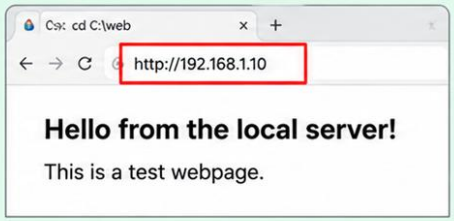
## Wireshark Demonstration: Capture HTTP Traffic on a Local Test Server

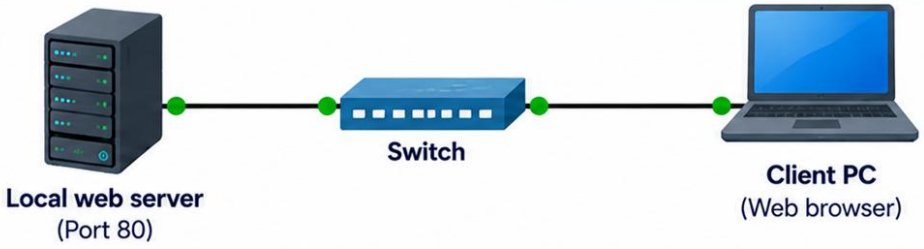
Inspect a GET request and server response

**1 Start a local web server**
(e.g. Python SimpleHTTPServer)

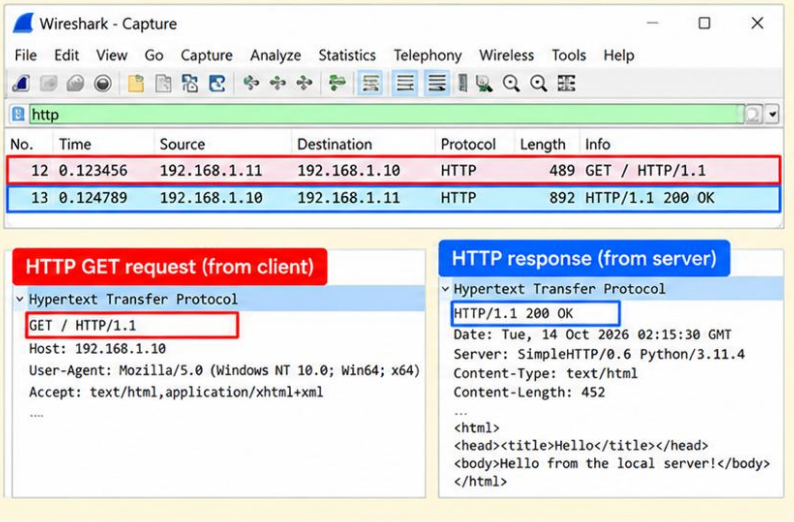



**2 Open a webpage in a browser**
(e.g. http://192.168.1.10)




**3 Capture and inspect HTTP traffic in Wireshark**
Find the GET request and the server response.






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# How Does Email Travel Across the Internet?

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Email uses different protocols for sending and accessing messages.

SMTP (Simple Mail Transfer Protocol) sends and relays emails.

IMAP (Internet Message Access Protocol) allows users to access and synchronise emails across devices.

POP3 (Post Office Protocol version 3) downloads emails from a mail server.


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# How Does Email Travel Across the Internet?

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Example in Australia:

A lecturer sends an email from Sydney to a student in Melbourne.

1. The lecturer writes and sends an email.
2. SMTP submits the message to the outgoing mail server.
3. Mail servers transfer the email towards the recipient's server.
4. The student accesses the email using a mail application or webmail.

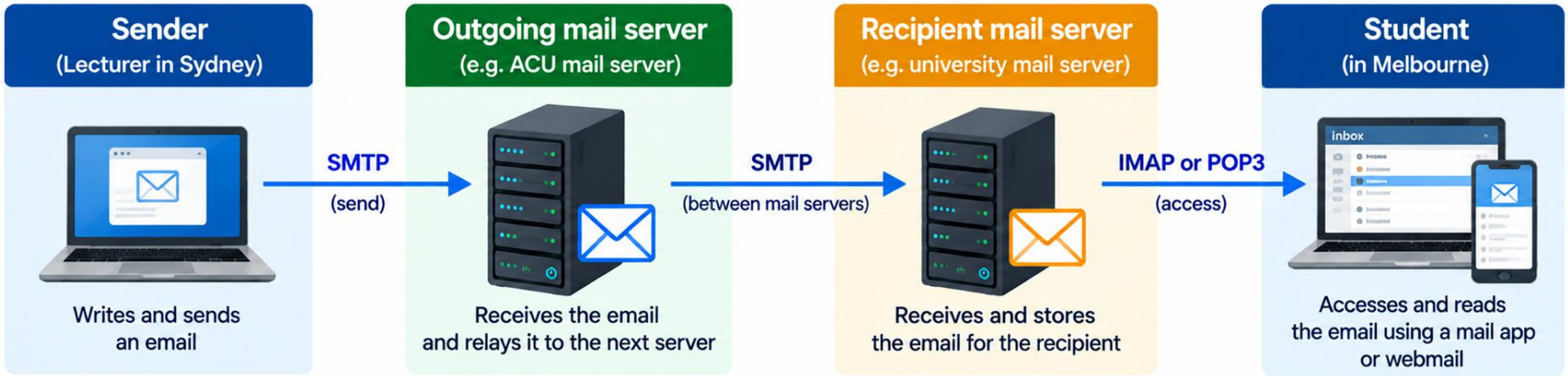
**Discussion question:** Which protocol is responsible for sending an email between mail servers?

Simple answer: SMTP.


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# How Does Email Travel Across the Internet?

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# SMTP, POP3 and IMAP: Ports and Differences

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| Protocol | Common ports | Main purpose |
|---|---|---|
| SMTP | 25, 587, 465 | Sending and relaying emails |
| POP3 | 110, 995 | Downloading emails |
| IMAP | 143, 993 | Accessing and synchronising emails |


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# SMTP, POP3 and IMAP: Ports and Differences

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Important concepts:

- SMTP port 25 is commonly used for mail-server delivery.
- Port 587 is commonly used for authenticated email submission.
- Ports 465, 995 and 993 are commonly used with TLS encryption.
- POP3 can download messages and optionally remove them from the server.
- IMAP generally keeps mail synchronised across devices.


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# SMTP, POP3 and IMAP: Ports and Differences

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Example in Australia: A university staff member checks the same mailbox using a laptop at work and a phone while travelling. IMAP allows both devices to access the same server-side mailbox.

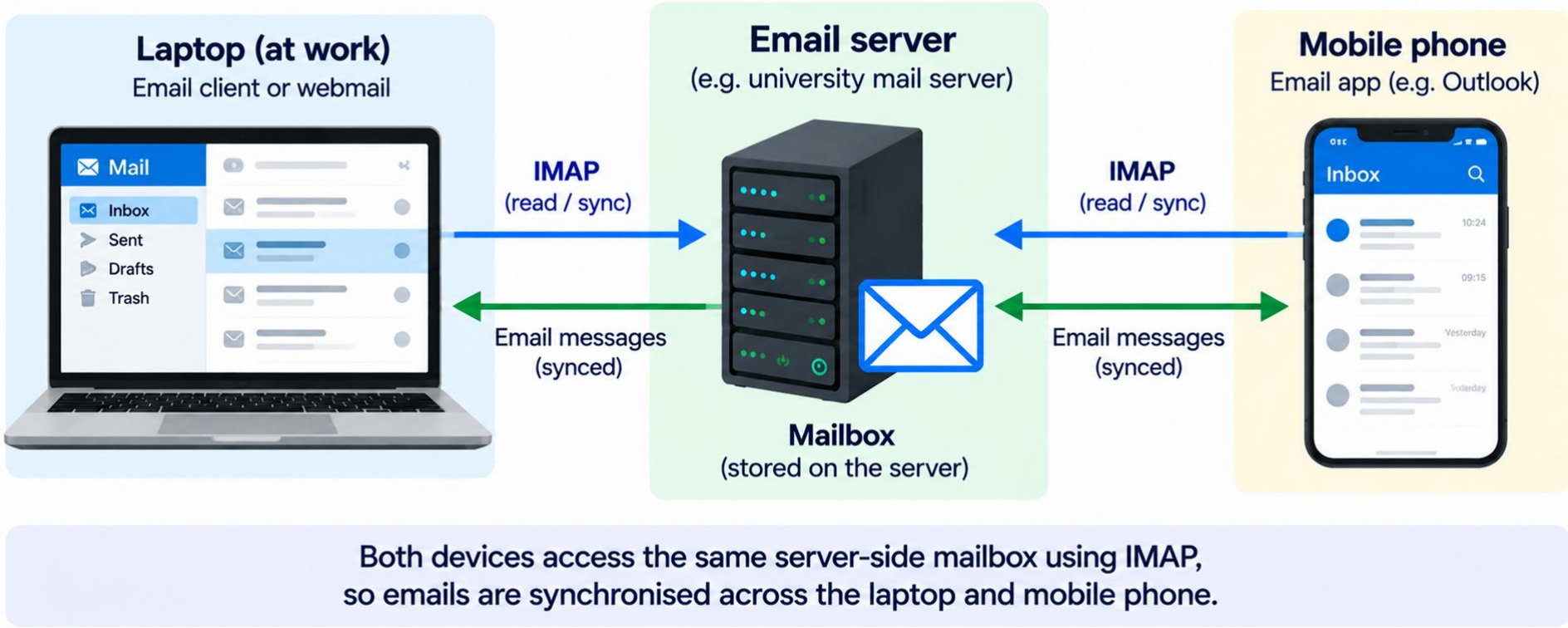
**Discussion question**: Which protocol is generally better when checking the same mailbox from a phone and laptop?

Simple answer: IMAP, because it synchronises messages across devices.


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# SMTP, POP3 and IMAP: Ports and Differences

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Both devices access the same server-side mailbox using IMAP, so emails are synchronised across the laptop and mobile phone.



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# DNS: From Website Name to IP Address

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DNS (Domain Name System) helps computers find the IP addresses associated with domain names.

How it works:

1. A user enters a website name.
2. The computer checks for an existing DNS result.
3. If needed, a DNS resolver looks up the name.
4. The resolver returns the relevant IP address.
5. The browser can then connect to the web service.


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# DNS: From Website Name to IP Address

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Example in Australia:

A student types www.abc.net.au.

DNS helps the browser find an address for the website.

Discussion question: Does DNS download the webpage?

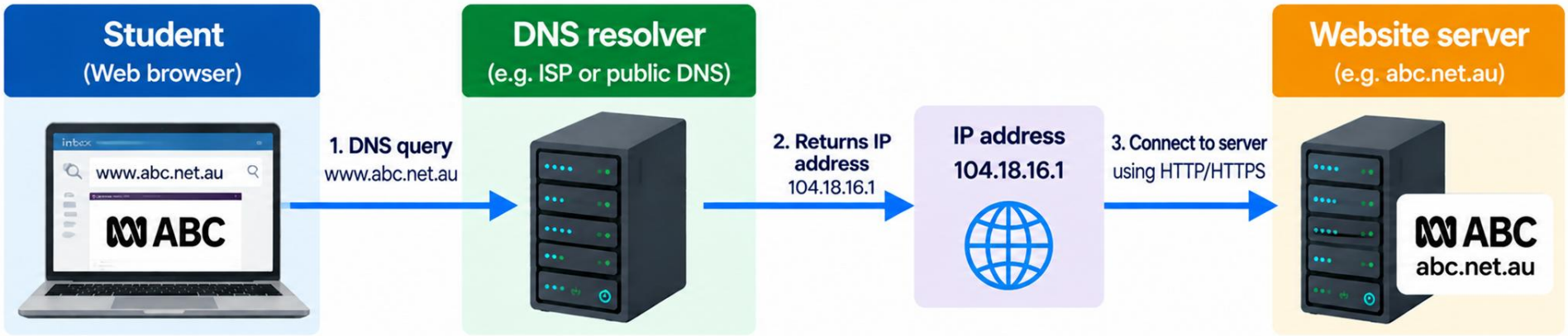
Simple answer: No. DNS helps find the server's address.

HTTP or HTTPS is then used to request web content.


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# DNS: From Website Name to IP Address

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1. The student types **www.abc.net.au** into the browser.
2. The DNS resolver looks up the domain name and finds the corresponding IP address.
3. The DNS resolver returns the IP address (e.g. **104.18.16.1**) to the student's device.
4. The student's browser connects to the website server using the IP address and then requests the webpage using HTTP or HTTPS.



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# Understanding the DNS Hierarchy

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DNS is organised into a hierarchy.

| Level | Example | Meaning |
|---|---|---|
| Root | . | Top of the DNS hierarchy |
| Top-level domain | .au | Australian country-code domain |
| Second-level domain | com.au | Domain category within .au |
| Registered domain | example.com.au | Organisation's registered name |
| Hostname | www.example.com.au | Specific name within the domain |


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# Understanding the DNS Hierarchy

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How DNS servers work together:

- Root servers help direct lookups to top-level domain servers.
- Top-level domain servers direct lookups towards the relevant authoritative servers.
- Authoritative DNS servers provide records for their domains.
- Recursive resolvers perform lookups and cache results for clients.


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# Understanding the DNS Hierarchy

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Example in Australia: The domain www.abc.net.au uses the Australian .au namespace, with net.au beneath it.

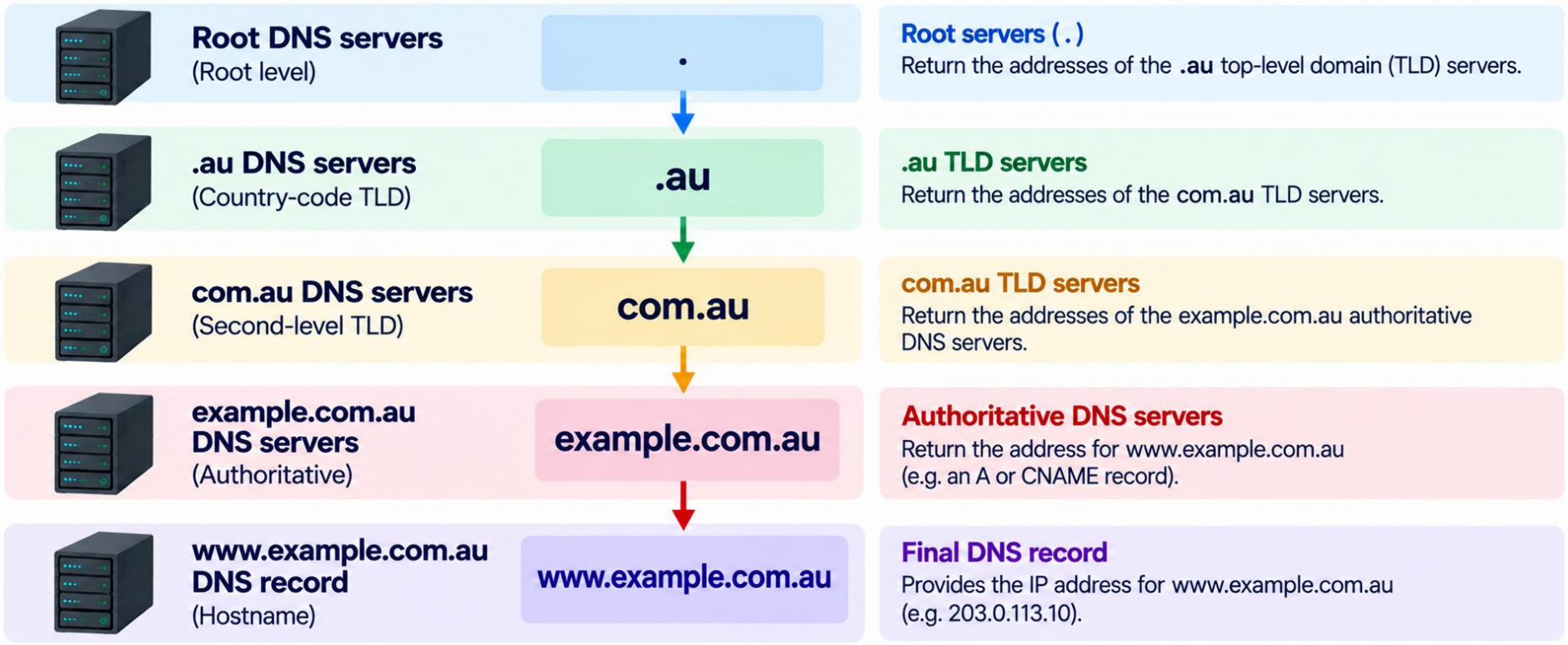
**Discussion question:** What does .au represent in a domain name?

Simple answer: Australia's country-code top-level domain.


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# Understanding the DNS Hierarchy

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# Using nslookup to Find IP Addresses

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nslookup is a command-line tool used to query DNS servers.

It can help network administrators:

- Find IP addresses for domain names.
- Check DNS records.
- Identify DNS configuration problems.
- Troubleshoot website access.


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# Using nslookup to Find IP Addresses

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Example in Australia:

Open Windows Command Prompt and enter:

nslookup www.abc.net.au

The command displays the DNS server used and the IP address or addresses returned.

Discussion question: If a website does not open, why might a network technician use nslookup?

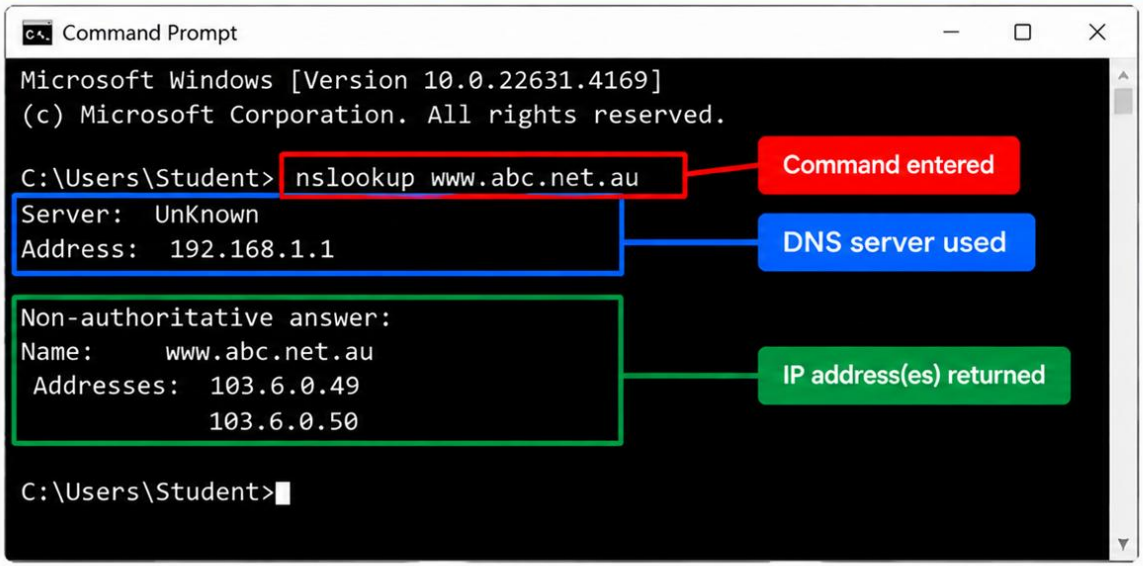
Simple answer: To check whether DNS can find an IP address for the website.


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# Using nslookup to Find IP Addresses

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## Windows Command Prompt: nslookup www.abc.net.au




**1 Run the command**

`nslookup www.abc.net.au`

This queries DNS to find the IP address of the website.

**2 Check the DNS server**

The output shows the DNS server used for the lookup (e.g. 192.168.1.1, or a DNS server provided by your ISP).

**3 View the IP address(es)**

The output shows the IP address or addresses for www.abc.net.au (e.g. 103.6.0.49, 103.6.0.50). A website may have multiple IP addresses.

**What does this show?**

The nslookup command confirms that DNS can find the IP address for www.abc.net.au, which is then used by your browser to connect to the website.


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# DNS Troubleshooting: A Practical Example

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Scenario: A student can access a website using its IP address but cannot access it using its domain name.

Possible explanation: A DNS resolution problem.

Troubleshooting steps:

1. Check the network connection.
2. Check the configured DNS server.
3. Run nslookup to test name resolution.
4. Compare results using another DNS resolver if permitted.
5. Investigate incorrect DNS settings or unavailable DNS services.


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# DNS Troubleshooting: A Practical Example

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Example in Australia: A student in a university computer lab cannot open a learning website by name, although other network connectivity tests work.

Discussion question: If an IP address works but a domain name fails, which service should you investigate first?

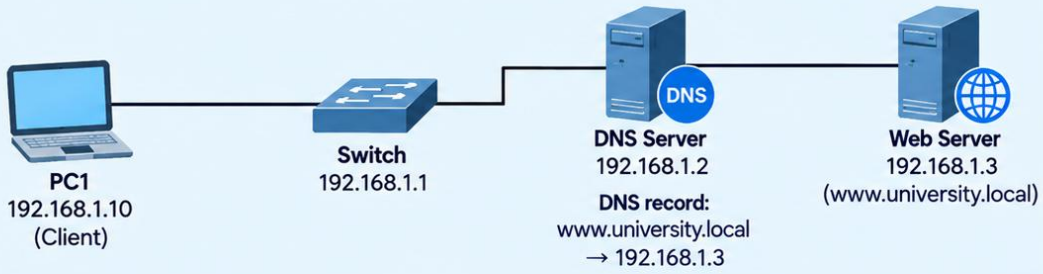
Simple answer: DNS, because it is responsible for translating domain names into addresses.


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# DNS Troubleshooting: A Practical Example

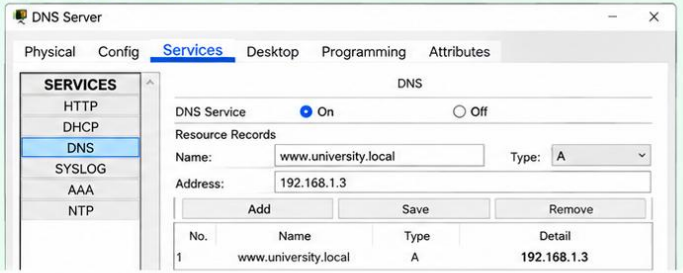
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## Cisco Packet Tracer – DNS Server Demonstration




### 1 Configure DNS server

Add an A record for www.university.local pointing to 192.168.1.3.




### 2 Test name resolution (working)

PC1 is configured to use the DNS server (192.168.1.2).

```
PC1
C:\>nslookup www.university.local
Server:    192.168.1 .2
Address:   192.168.1 .2

Name:     www.university.local
Address:  192.168.1.3

C:\>
```

Name resolution successful (IP address returned)

### 3 Access the website (working)

PC1 can open the website using the domain name.

```
PC1
C:\>ping www.university.local
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
C:\>
```

### 4 Change DNS configuration (failure)

Set PC1 to use an incorrect DNS server (e.g. 8.8.8.8 or a non-existent address).

```
PC1 (DNS changed)
C:\>nslookup www.university.local
Server:    8.8.8.8
Address:   8.8.8.8
*** 8.8.8.8 can't find www.university.local:
Non-existent domain

C:\>
```

Name resolution fails (no IP address returned)


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# What Is DHCP?

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DHCP (Dynamic Host Configuration Protocol) automatically provides network configuration to devices.

DHCP can provide:

- IP address
- Subnet mask
- Default gateway
- DNS server address


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# What Is DHCP?

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Important terms:

- DHCP server: Provides network configuration.
- DHCP client: Requests configuration.
- Address pool: A range of addresses available for assignment.
- Lease: The period for which an IP address is assigned.

Example in Australia: When a student connects a laptop to a university Wi-Fi network, DHCP commonly provides the device with an IP address and other network settings.


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# What Is DHCP?

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**Discussion question:** Why is DHCP useful in a university with thousands of devices?

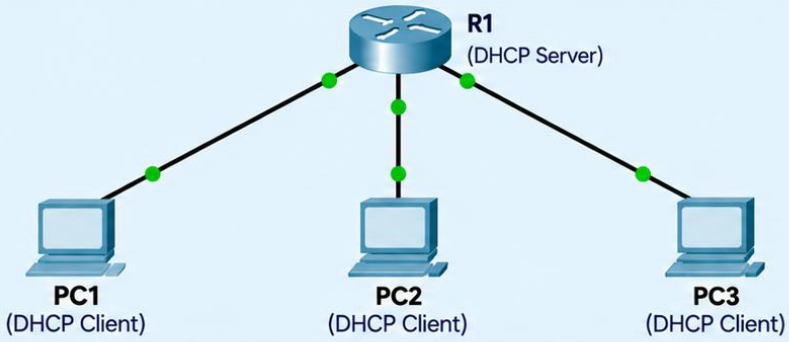
Simple answer: It automatically configures devices and reduces the need to enter IP addresses manually.


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# What Is DHCP?

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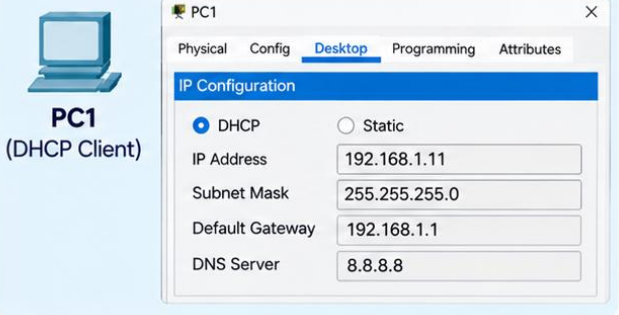
## 1 Network topology (Cisco Packet Tracer)

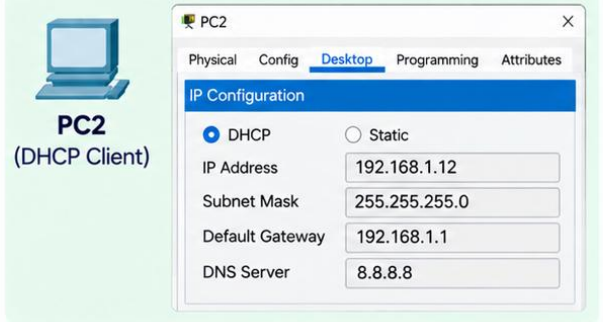


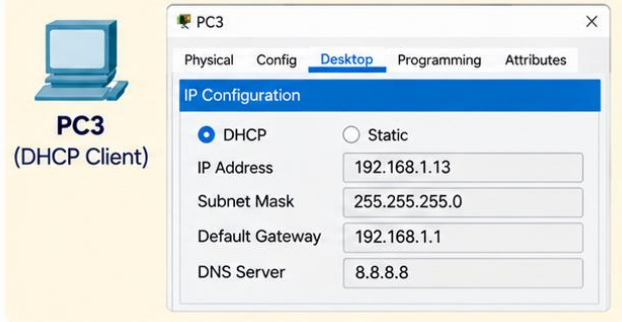

## 2 R1 DHCP configuration (CLI)

```
R1> enable
R1# configure terminal
R1(config)# ip dhcp excluded-address  192.168.1.1 192.168.1.10
R1(config)# ip dhcp pool LAN
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.1.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# lease 7
R1(dhcp-config)# exit
R1(config)# end
R1# show ip dhcp binding
```

## 3 PCs receive IP addresses from DHCP



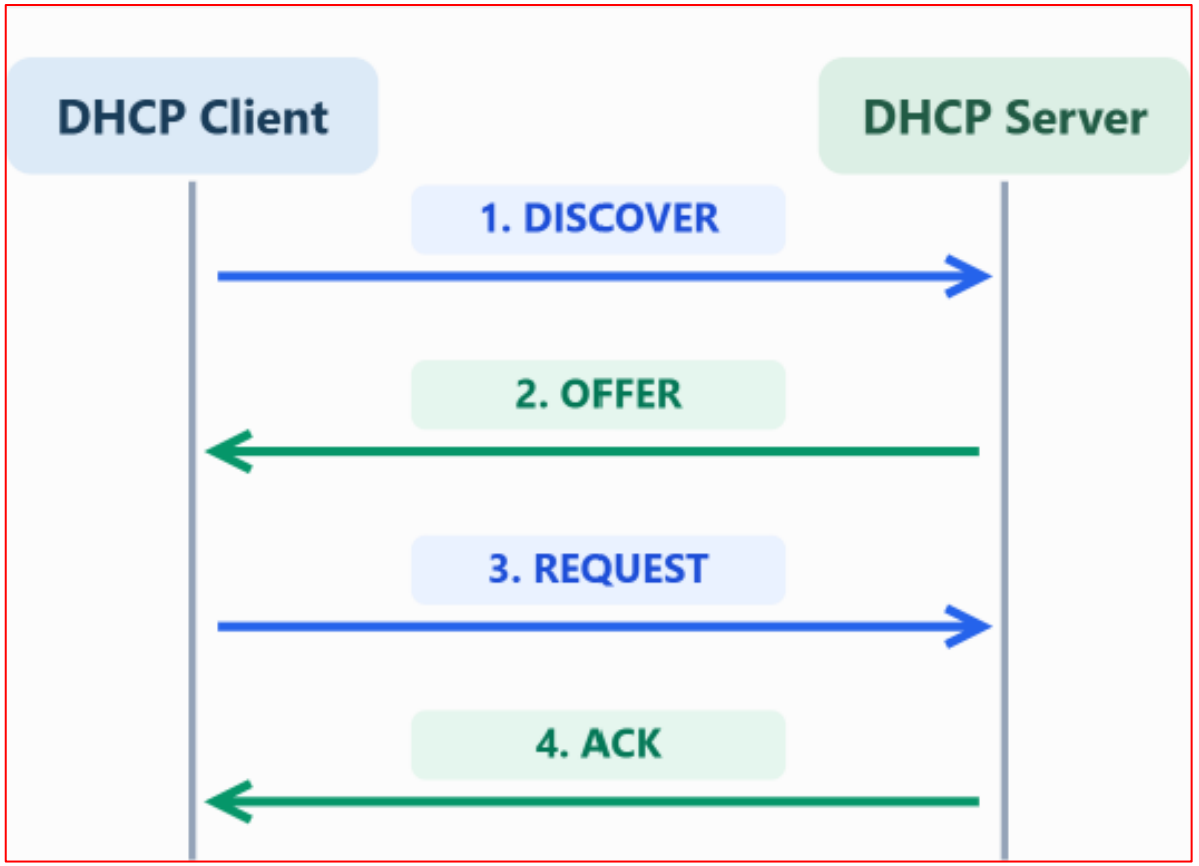


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# How DHCP Works: The DORA Process

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DHCP commonly uses four messages to assign an IPv4 address.





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# How DHCP Works: The DORA Process

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DORA means:

1. Discover: The client searches for a DHCP server.
2. Offer: A DHCP server offers an IP address and configuration.
3. Request: The client requests the offered configuration.
4. Acknowledge: The server confirms the lease.

Example in Australia: When a customer connects a new laptop to a home Wi-Fi network in Sydney, the router commonly completes the DHCP process automatically.


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# How DHCP Works: The DORA Process

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Discussion question: Which DHCP message confirms that the IP address lease has been assigned?

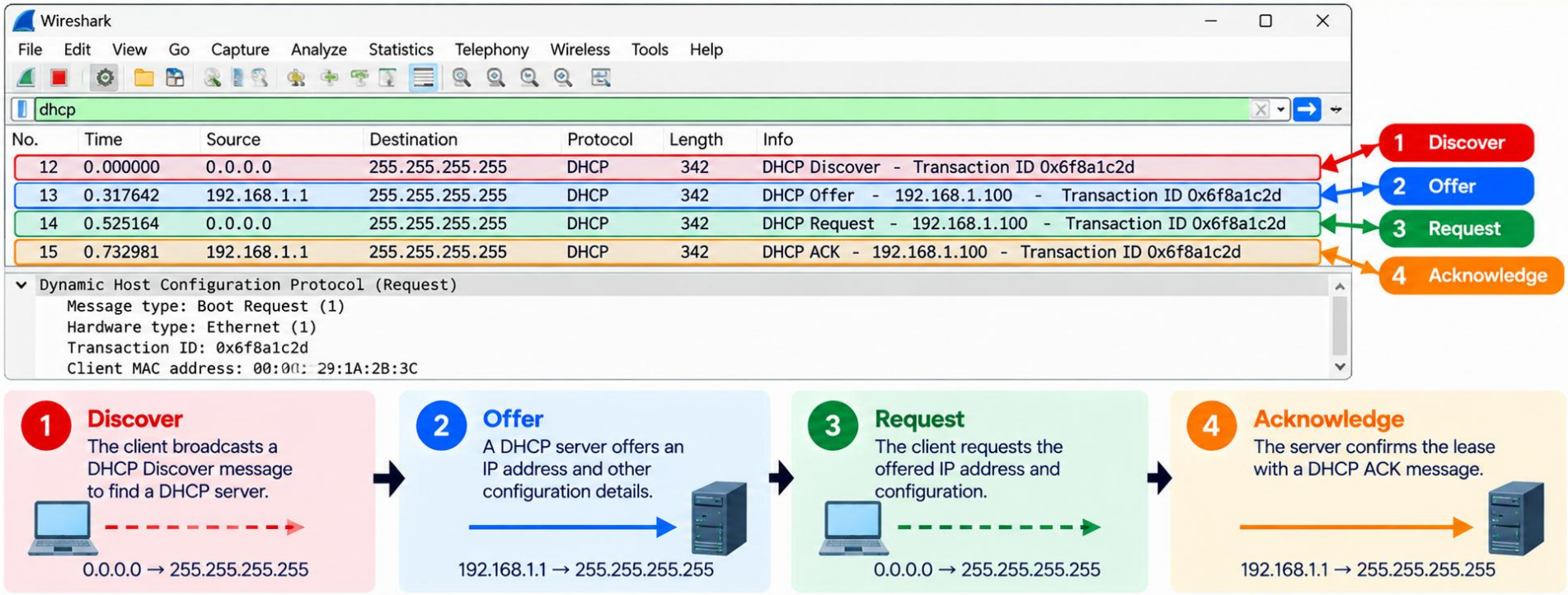
Simple answer: DHCP ACK (Acknowledgement).


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# How DHCP Works: The DORA Process

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## Wireshark – Capture and analyse DHCP traffic (DORA process)




**1 Discover**
The client broadcasts a DHCP Discover message to find a DHCP server.

0.0.0.0 → 255.255.255.255

**2 Offer**
A DHCP server offers an IP address and other configuration details.

192.168.1.1 → 255.255.255.255

**3 Request**
The client requests the offered IP address and configuration.

0.0.0.0 → 255.255.255.255

**4 Acknowledge**
The server confirms the lease with a DHCP ACK message.

192.168.1.1 → 255.255.255.255


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# Static vs Dynamic IP Addressing

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There are two common ways to assign IP addresses.

| **Feature** | **Static addressing** | **Dynamic addressing (DHCP)** |
|---|---|---|
| Assignment | Manually configured | Automatically assigned |
| Changes | Usually remains fixed | Can change when a lease changes |
| Administration | Requires manual configuration | Easier for large numbers of devices |
| Typical use | Routers, servers, network infrastructure | Student laptops, phones, guest devices |


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# Static vs Dynamic IP Addressing

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Important: Servers and printers can also use DHCP reservations to receive predictable addresses without manually configuring each device.

Example in Australia:

A university may use:

- Static addresses or DHCP reservations for servers and printers.
- DHCP for student laptops and mobile devices.


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# Static vs Dynamic IP Addressing

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**Discussion question:** Would you manually configure IP addresses for 500 student laptops?

Simple answer: Usually not. DHCP is more practical because it automatically assigns network settings.


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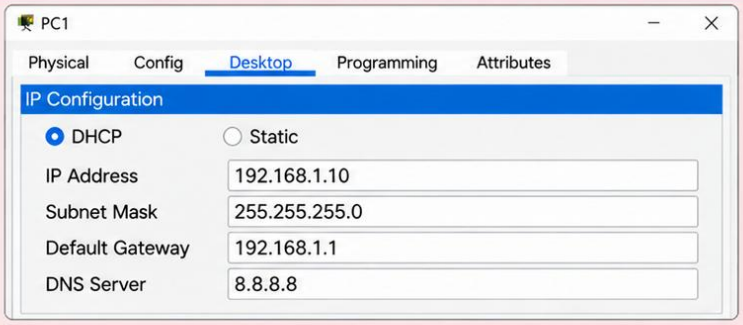
# Static vs Dynamic IP Addressing

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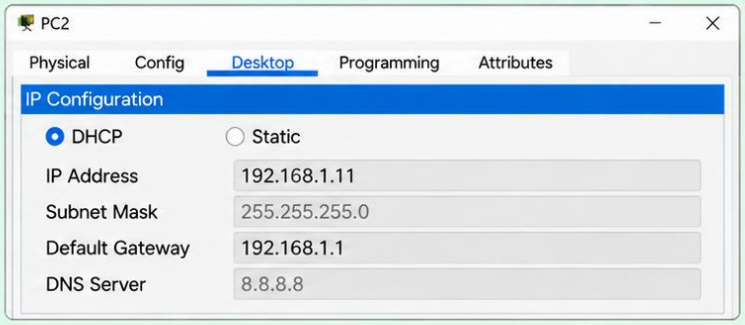
## Cisco Packet Tracer – Compare Static IP and DHCP




### 1 PC1 – Manually configured static IP address




### 2 PC2 – Obtaining settings through DHCP




| PC | IP configuration method | IP address | Subnet mask | Default gateway | DNS server |
|---|---|---|---|---|---|
| **PC1 (Static)** | Manual (Static IP) | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 | 8.8.8.8 |
| **PC2 (DHCP)** | Automatic (DHCP) | 192.168.1.11 (assigned) | 255.255.255.0 | 192.168.1.1 | 8.8.8.8 |


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# FTP: Transferring Files Between Computers

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FTP (File Transfer Protocol) allows files to be uploaded and downloaded between a client and a server.

- FTP client: The application used to send or receive files.
- FTP server: The computer or service that provides access to files.
- Upload: Sending a file from your computer to a server.
- Download: Receiving a file from a server onto your computer.


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# FTP: Transferring Files Between Computers

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FTP uses TCP and normally establishes two connections:

- Control connection (TCP port 21): Sends commands and receives replies.
- Data connection: Transfers files and directory listings using a separate connection.

Traditional FTP does not encrypt its traffic. Secure alternatives include SFTP and FTPS.


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# FTP: Transferring Files Between Computers

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Example in Australia: A university IT administrator transfers website files from a workstation to a university web server using a secure file-transfer service.

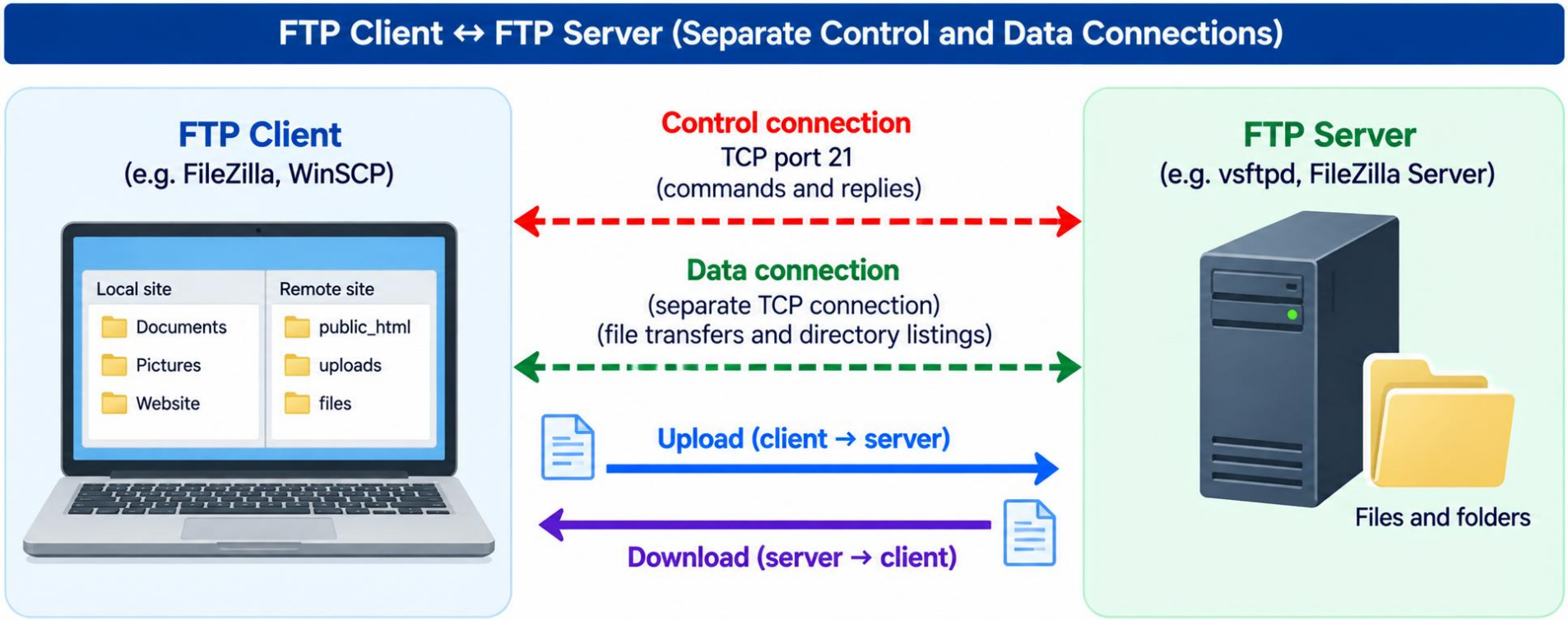
Discussion question: What is the difference between uploading and downloading a file?

Simple answer: Uploading sends a file to a server; downloading receives a file from a server.


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# FTP: Transferring Files Between Computers

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# FTP vs SFTP: Why Security Matters

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Traditional FTP and SFTP can both transfer files, but they work differently.

| Feature | FTP | SFTP |
|---|---|---|
| Main purpose | File transfer | Secure file transfer |
| Encryption | No encryption by default | Encrypted using SSH |
| Common control/service port | TCP 21 | TCP 22 |
| Connections | Separate control and data connections | Normally one SSH connection |
| Recommended for sensitive files | No | Yes |


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# FTP vs SFTP: Why Security Matters

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Important: SFTP is not simply FTP with encryption added. It is a separate file-transfer protocol operating over SSH.

Example in Australia: An Australian organisation transferring confidential business reports should use an appropriately secured transfer service rather than traditional unencrypted FTP.

Discussion question: Why is SFTP safer than traditional FTP?

Simple answer: SFTP encrypts the connection, helping protect files and login information during transfer.

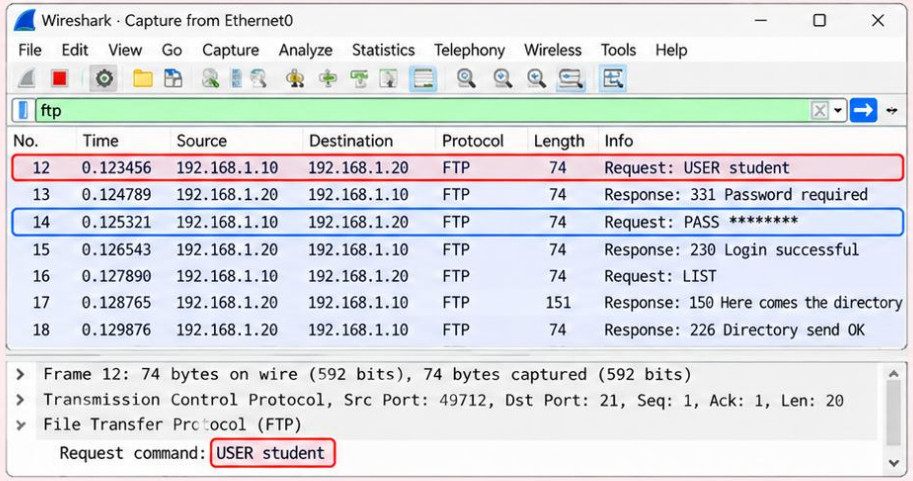

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# FTP vs SFTP: Why Security Matters

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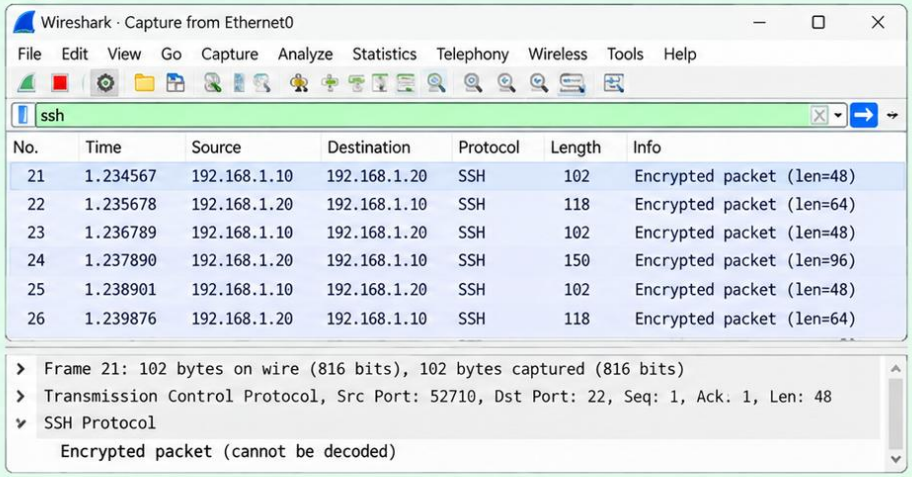
## Wireshark – Compare FTP commands with encrypted SSH/SFTP traffic

### 1 FTP traffic (commands visible)




- **FTP** uses plain text, so commands (e.g. USER, PASS, LIST) are visible in Wireshark.
- Login information and file commands can be read by others on the network.
- Traditional FTP does not encrypt its traffic.

### 2 SFTP traffic (encrypted)




- **SFTP** runs over SSH (typically TCP port 22).
- Traffic is encrypted, so file commands and data are not visible in Wireshark.
- This helps protect login information and files during transfer.


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# SMB: Sharing Files and Printers

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SMB (Server Message Block) allows computers to access shared files, folders and printers over a network.

- SMB commonly uses TCP port 445.
- A server makes a folder or printer available.
- An authorised client connects to the shared resource.
- The client can access the shared resource through its operating system.
- Access permissions determine what users can view or change.


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# SMB: Sharing Files and Printers

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Example in Australia: Computers in a university laboratory access a shared network folder containing teaching materials, or use a shared printer managed by the IT department.

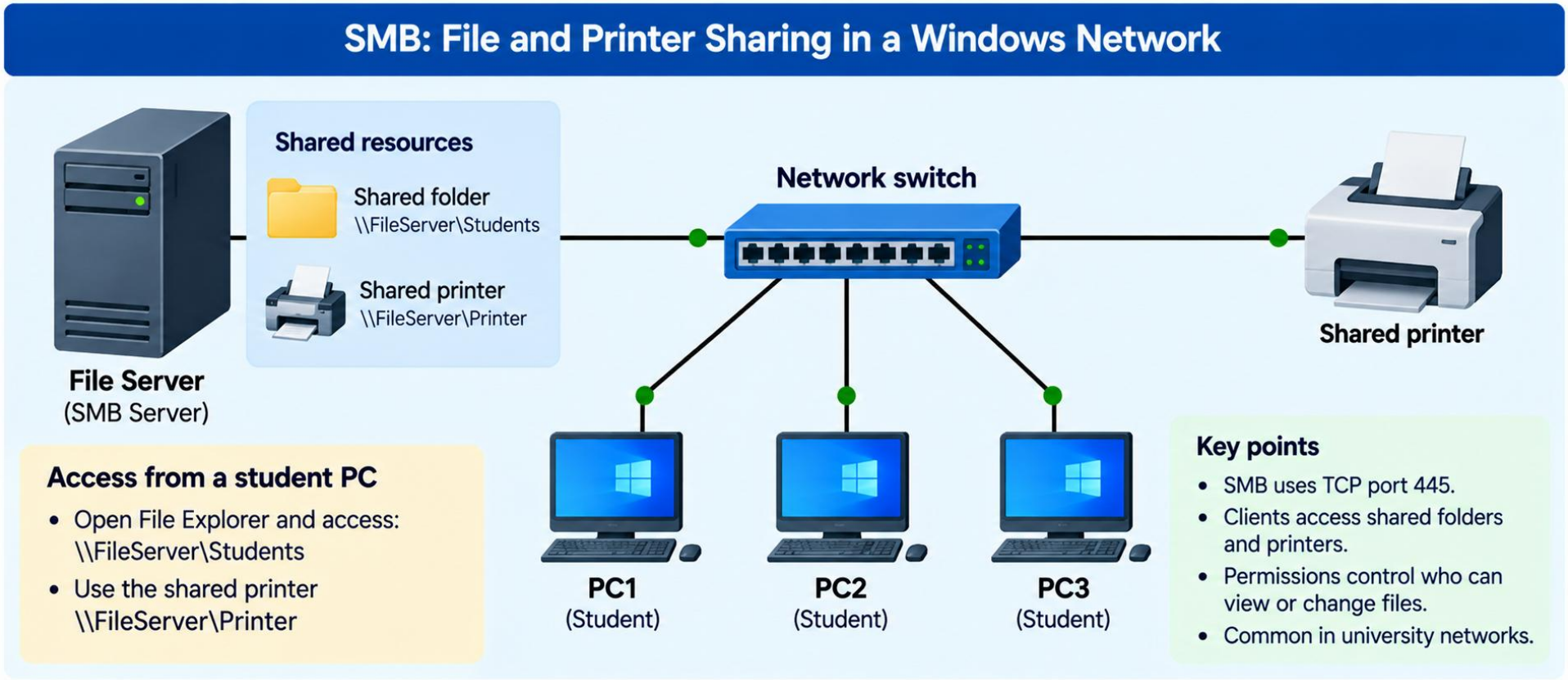
Discussion question: If 30 computers need to use one shared printer, which protocol might support the file or printer-sharing connection in a Windows network?

Simple answer: SMB.


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# SMB: Sharing Files and Printers

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# How Data Travels Through Network Layers

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When you open a website, data passes through several networking layers.

This process is called encapsulation.

| Layer | What happens | Data unit |
|---|---|---|
| Application | Creates an HTTP request | Application data |
| Transport | Adds TCP information, including port numbers | TCP segment |
| Internet/Network | Adds source and destination IP addresses | IP packet |
| Network Access | Adds local network framing information | Frame |
| Physical transmission | Sends the information over a physical medium | Bits/signals |


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# How Data Travels Through Network Layers

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At the receiving device, the process is reversed. This is called decapsulation.

Example in Australia: A student in Sydney opens a university learning website. The browser creates a request, TCP carries the data, IP helps route it across networks, and Ethernet or Wi-Fi carries it across the local network.

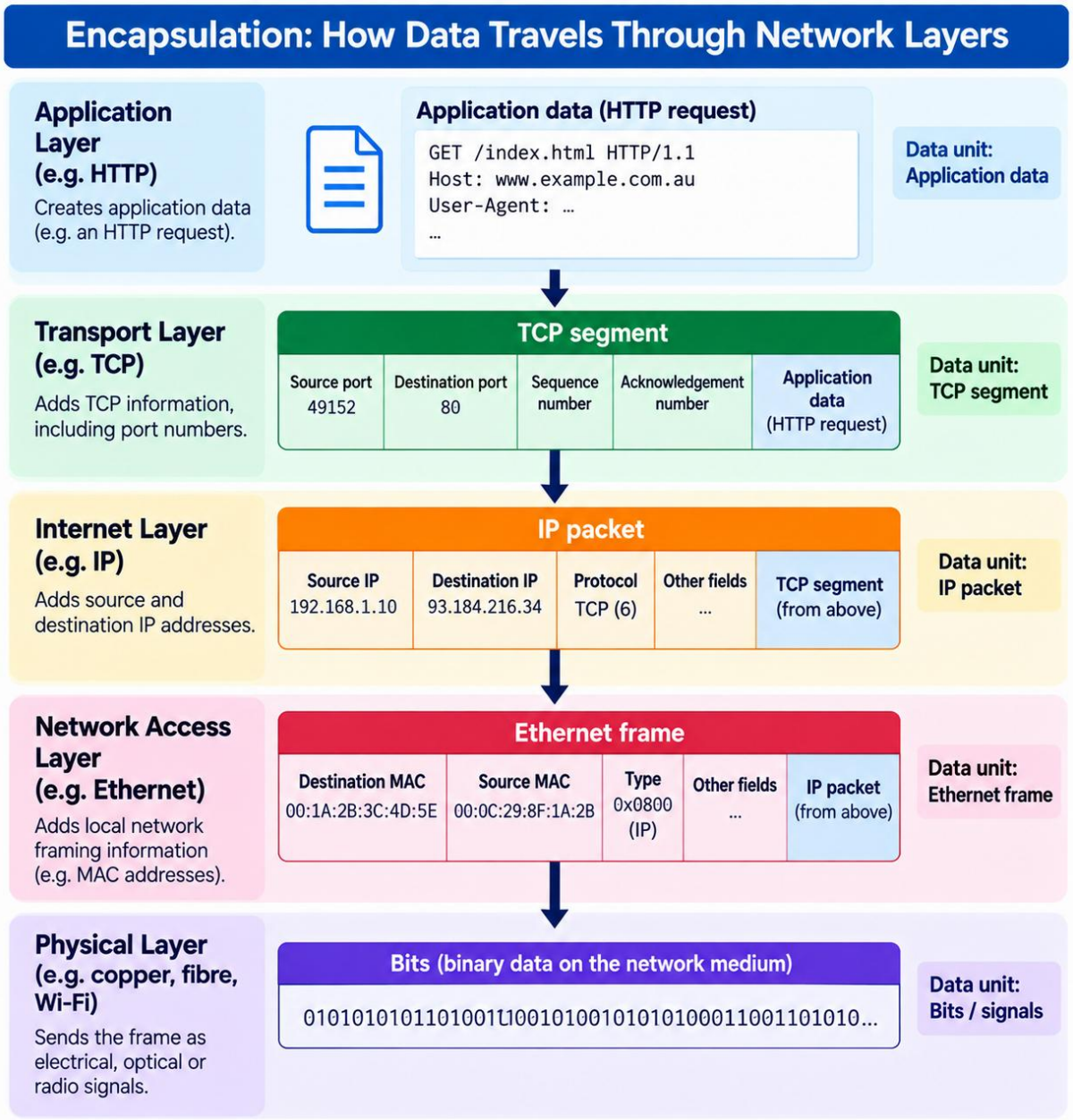
Discussion question: Which layer adds the source and destination IP addresses?

Simple answer: The Internet (Network) layer.


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# How Data Travels Through Network Layers

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# Port Numbers: Delivering Data to the Correct Application

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A computer can run many network applications at the same time.

Port numbers help the transport layer direct incoming data to the correct application or service.

For example:


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# Port Numbers: Delivering Data to the Correct Application

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| Service | Common port | Transport |
|---|---|---|
| HTTP | 80 | TCP |
| HTTPS | 443 | TCP; also UDP for HTTP/3 |
| FTP control | 21 | TCP |
| SSH/SFTP | 22 | TCP |
| SMTP | 25 | TCP |
| DNS | 53 | UDP or TCP |
| SMB | 445 | TCP |


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# Port Numbers: Delivering Data to the Correct Application

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Example in Australia: A student may open a university website, check email and access a shared folder simultaneously. Different services and connections can be identified using port numbers and other connection information.

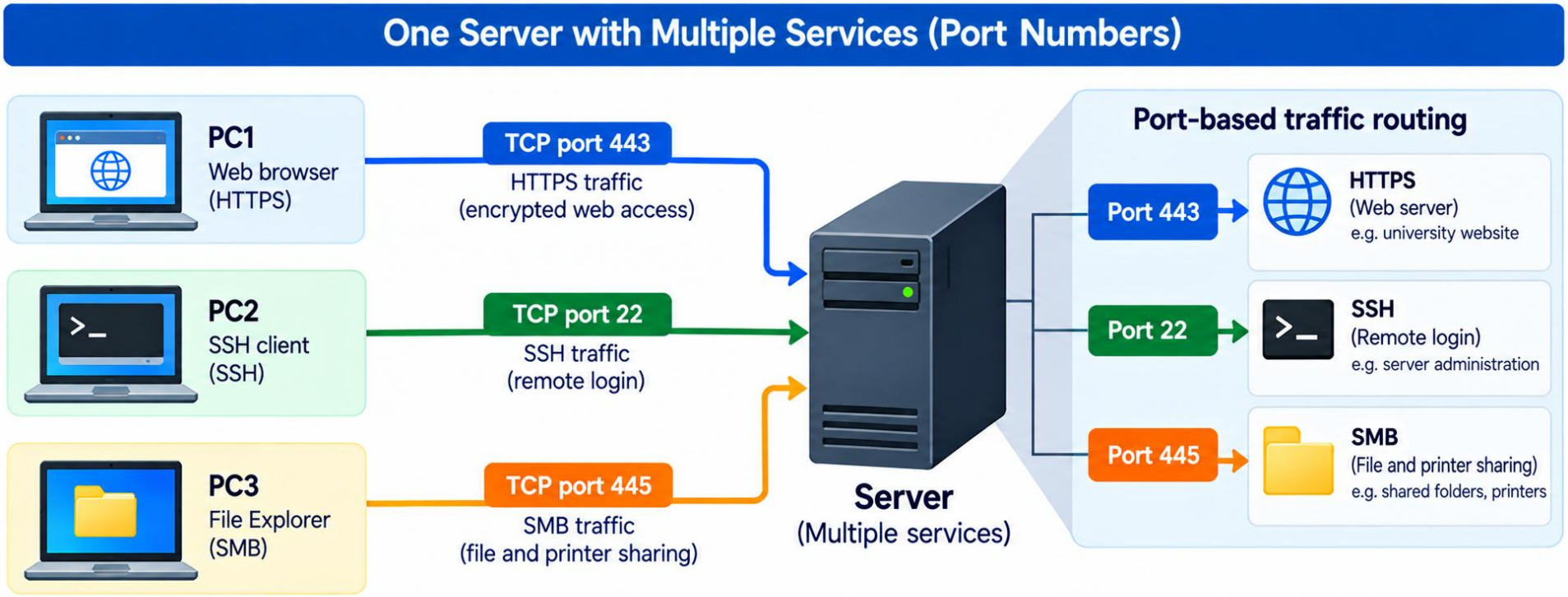
**Discussion question:** A server receives traffic for TCP port 445. Which service is commonly associated with this port? Simple answer: SMB, used for network file and printer sharing.



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# Port Numbers: Delivering Data to the Correct Application

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# Week 10 Summary: Application Layer Protocols

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The application layer provides network services used by applications such as browsers, email clients and file-sharing software.

| Protocol | Main purpose | Everyday example |
|---|---|---|
| HTTP/HTTPS | Access web content | University website |
| SMTP | Send email | Sending an assignment-related email |
| IMAP/POP3 | Access email | Checking a university mailbox |
| DNS | Find IP addresses from names | Opening an Australian website |
| DHCP | Obtain network settings automatically | Connecting to campus Wi-Fi |
| FTP/SFTP | Transfer files | Uploading files to a server |
| SMB | Share files and printers | Accessing a laboratory shared folder |

# Week 10 Summary: Application Layer Protocols

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Key message: These protocols provide different services, but they work together with the transport, network and network-access layers.

**Discussion question:** When you connect a laptop to Wi-Fi and open a university website, which two application-layer services might be used before the webpage loads?

Simple answer: DHCP may provide network settings, and DNS may find the website's IP address. HTTPS can then retrieve the webpage.


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# Let's Answer Your Questions

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- *Education is the most powerful weapon you can use to change the world.* — Nelson Mandela